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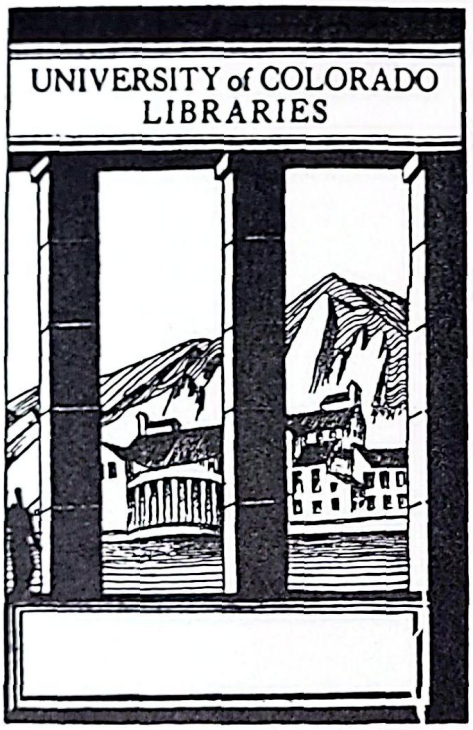
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AN EXPERIMENTAL INVESTIGATION OF THE RELATION OF MOTIVATION TO THE DEGREE OF SPATIAL LEARNING IN THE WHITE RAT.

by

Helen Chamberlain

B.A., State University of Iowa, 1943

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A Thesis submitted to the Faculty of the Graduate School of the University of Colorado in partial fulfillment of the requirements for the Degree

Master of Arts

Department of Psychology

1947

This Thesis for the M.A. degree, by

Helen Chamberlain

has been approved for the

Department of

Psychology

by

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Chamberlain, Helen (M. A., Psychology)

An Experimental Investigation of the Relation of Motivation to the Degree of Spatial Learning in the White Rat.

Thesis directed by H. H. Kendler

Recent work by Tolman and his co-workers has shown that animals can make a correct choice of a new path leading to a goal when the original path to the goal is blocked. This behavior has been called spatial learning. The present study was an attempt to investigate the effect of varying degrees of motivation on the amount of spatial learning exhibited.

An apparatus was used which permitted a period of training trials on a T-maze and a period of test trials involving the choice of one of six pathways, one of which led directly to the goal box which had contained food in the training trials. The two motivational states used were 6 and 21 hours of food deprivation. The deprivation periods were changed in two of the groups between the training and test series, whereas in the other two groups they were held constant. Two experiments were performed, one in which a light was placed directly over the choice point of the T-maze, and the other in which the light was placed directly over the rewarding goal box.

The results indicate that the degree of motivation

had no effect upon the amount of spatial learning exhibited by the subjects. It was also found that the animals in the experiment in which the light was directly over the goal box gave more evidence of spatial learning than did those in the experiment in which the light was over the choice point in the T-maze. The degree of spatial learning on the first test trial also varied from that found by Tolman and his co-workers.

The results were analyzed within the theoretical framework of both Hull's and Tolman's theories of learning.

This abstract of about 250 words is approved as to form and content. I recommend its publication.

Signed Howard H. Kendler

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INTRODUCTION

Recently, Tolman, Ritchie, and Kalish (28,36,37,38) have published several of a series of experimental articles entitled "Studies in Spatial Learning." In these studies the authors have attempted to specify more rigidly the implications of Tolman's theory of learning and to provide pertinent empirical evidence in support of Tolman's newer formulation of his theory of expectancy.

In this theory Tolman has assumed that what is learned is a "cognitive map" which is representative of certain features of the environment. One important feature of such a "map" is the location of the reward. Thus, food is "expected" by the animal in a certain location and the animal, under relevant motivational conditions, tends to make the appropriate response toward the place of food. This theory is in contrast to theories such as those proposed by Hull (19) and Guthrie (13) who assert that what is learned is a response to certain stimuli. Because of the nature of their theoretical organization, these theorists have concerned themselves with those experimental situations in which the response learned can be objectively and precisely measured as it develops. Conditioning and simple trial-and-error learning situations admirably fulfill these requirements. Since these theories are concerned with the development of S-R asso-

association in which the response is more or less constant, it appears appropriate to refer to them as "response-reproduction theories."

In the initial article (36), which is the one most directly related to the present study, Tolman, Ritchie, and Kalish have attempted to differentiate between the response-reproduction theories and Tolman's formulation of place-learning. They state that it has been difficult to distinguish between the response-reproduction theories and place-learning theory because of an apparent similarity in concepts despite marked differences in terminology. They illustrate this apparent similarity by re-stating Tolman's formulation of the theory of expectancy which, although never stated in this exact form, was implicit in the earlier formulation. They write:

> When we assert that a rat expects food at location L, what we assert is that if (1) the rat has been deprived of food for more than twelve hours, (2) he has been trained on path P, and (3) he is now placed on path P, then he will run down path P. (36,p.14)

Commenting on this formulation Tolman, Ritchie, and Kalish point out that:

> ...the differences between the stimulus-response and expectancy theories, when "expectation" is defined in such a fashion, are purely terminological. What we would call "signs," they would call "stimuli," and what they would call "response-tendencies," we would call "expectations." (36,p.14-15)

In an attempt to differentiate between these two theories, they present a revised formulation:

> When we assert that a rat expects food at location L, what we assert is that if (1) he is deprived of food, (2) he has been trained on path P, (3) he is now put on path P, (4) path P is now blocked, and (5) there are other paths which lead away from path P, one of which points directly to location L, then he will run down the path which points directly to location L. (36,p.15)

Therefore, Tolman would say, that the rat does not learn in a maze situation to make the correct response, but rather learns a cognitive map of the situation, of which one of the principal features is the place of reward. Thus, when the original path to the goal is blocked, the animal is able to cope with the new situation by taking the alternative route that will lead directly to the goal. Tolman calls this behavior spatial or place learning. It is here, in the prediction of appropriate new behavior, that Tolman and his co-workers feel that their theory is more adequate than those which assume that what is learned is a response.

HISTORY OF THE PROBLEM

Although Tolman's recent theorizing on the spatial nature of learning is the most articulate formulation of the concept, he is not the first worker to consider this point of view. Watson in 1907 (41) and Carr in 1917 (4), both reported that when a maze was rotated the behavior of the rat became confused and the errors increased. Hubbert and Lashley in 1917 (18) suggested that rats

in a maze soon acquire a sense of direction in relation to the goal. A similar conclusion was reached in 1920 by Hunter (21). He found, while working with double alternation problems, that rats could learn a spatial maze in which the alleys were arranged in llrrlrrll order, but that these same rats when transferred to a temporal maze could not learn the same alternation. He states:

> Running the spatial maze therefore, must require cues which have space location as well as temporal position. In other words the rat must recognize in terms of space where he is in the maze. (21,p.17)

Leuba and Fain (25) have also emphasized the importance of an orientation or reference point in the learning of a maze problem.

Lashley (24) observed that, when given the opportunity, rats would climb out of the starting box of the maze and run directly to the goal. Dennis (8) found, in an experiment designed to test the proprioceptive cues used by blinded animals in running a simple maze, that when the wall was removed:

> The rats simply passed over the position of the wall at the point where they were usually deflected by its presence. (8,p.70)

Nelson (15) reported a similar phenomenon. In his work on a discrimination problem, animals were shocked for making an incorrect response and then permitted to return to the choice point over the now uncharged grid in order to enter the correct goal box. He noted that instead of going over the formerly pain-producing grid, the animals

would blimb over the partition and into the rewarding goal box. Gengerelli (11) also found that rats tend to take the shortest route to food. In an open field offering an infinite number of paths to the goal box both blind and normal rats tended to take the most direct paths to the goal box.

Dashiell first observed the orientation of rats in 1920 (5), and later, in one of the most thorough studies of this behavior, (6) used a maze in which there were twenty alternate pathways of equal length from the entrance to the goal box. He discovered that the animals soon learned to run the maze but that they did not exhibit a strong perference for one pathway. They chosemany pathways and it was noted that it was common for an animal to run without error a particular pathway never before traversed. Dashiell accounted for this phenomenon by assuming that the rat, by experiencing the maze a number of times, developed a certain posturing or set toward the direction of the goal. It was this general direction orientation of the rat which made possible the "indiscriminate" choice and errorless running of the many alleys.

A novel situation was devised by Higginson (16) who forced rats to run past a closed door which, when open, led directly to the goal box. After the rat had run down a blind alley and back, the door to the goal box was opened and the animal allowed to enter. In the crucial trial the

door was left open at the start of the trial to test whether the animal would run past it and down the blind alley, as had been necessary in the previous trials; or whether it would change its behavior and take the correct path to the goal immediately. Five of nine rats did take the correct alley on the first trial.

Valentine (39) and Gilhousen (12) also turned their attention to the problem of the choice of a short cut to the goal in preference to the longer path on which the original training occured. Gilhousen observed that the short cut was not taken, and Valentine found that only a very small percentage of his animals chose the short cut alley on the first trial. Valentine attributed the choice of a short cut to the running speed of the rat as it passed the entrance to the new short-cut path.

A problem which has received a great deal of attention and is closely related to spatial learning is that of goal orinetation. This phenomenon is illustrated by the work of Tolman and Honzik (34,35). In these studies they found that in a maze situation the animal tends to make more errors in those blind alleys which point in the direction of the goal than in those blind alleys which point away from the goal. Warden and Fox (40) also studied this behavior. They trained rats to learn a maze and then when rotating the maze found that in the latter situation the animals tended on the first trial to make more errors on the alleys which pointed to the previous location of the goal. Ruch (29)

also reported similar goal direction or orientation.

Yoshioka has been one of the most diligent workers in this field. He found (44) that animals made the most errors on the alleys which pointed to the location of food. He also demonstrated goal orientation by finding (45) that rats ran along the wall of the maze pathway which was closer to the goal. He discovered (42,43) that the rat could distinguish between patterns in a maze which were of equal length to the goal but of different shapes. The apparatus he used (42) was a maze in the form of an equilateral triangle with alleys connecting the mid-points of both sides with the mid-point of the base. This left two choices of equal total length to the goal for the rat: the large equilateral triangle, or the two smaller triangles formed by the medians in the shape of a pentagon. The rats chose the pentagon more often than the large triangle. Tolman interprets this result,

> ...as due to the fact that the pentagon turns back toward the goal sooner than the triangle...the rats soon come to expect the general direction of the goal and that, given this expectation of the goal's direction, the pentagon seems to them better--to offer a more direct route to the goal than did the triangle. (33,p.124).

Dashiell and Bayroff (7) observed that rats learned a maze on which all the correct turns were on the side which pointed to the goal more easily than they learned a maze in which the correct alleys did not always point to the goal. They explain this by assuming a forward-going tendency in the animal which not only maintains for short

distances the direction already set, but leads to a "compensatory sort of correction" when the rat is forced out of line by an obstacle. This concept of a forward-going tendency in the rat is the result of an attempt to explain the facts of goal orientation by a hypothesis other than that which assumes that the animal somehow "knows" the direction of the goal.

Bayroff (2), in an experiment designed to determine whether the animal's behavior was dependent on direction orientation or on the forward-running behavior, found that the latter concept explained his results. Luh and Shen (27) agreed that the forward-going tendency was important but added an additional factor of the initial orientation in the entrance box.

Other theories to account for the making of more errors on the goal pointing alleys have been presented. Ballachey and Buel (1) attributed it to centrifugal swing. Buel (3) later concluded that the direction of the last turn to food was the most important factor. To test Buel's theory, Jones and Taylor (22) devised an experiment by which one group made the last turn in the direction of the goal and the other group did not. It would be predicted that the animals making the last turn away from the goal would make errors on the side that pointed to the goal. However, the two groups made the same number and type of errors. Jones and Taylor therefore conclude that:

> The position of the goal in space has had no mensurable effect on the maze behavior of the rats. (22,p.25)

In a review of the work done on goal orientation, Thompson (32) declared that goal orientation was just a "label", and as such it did not explain the behavior although it produced an impression of explanation. Therefore, he concludes, it is an unnecessary concept. Although such a comment is to some extent justified, the empirical results cannot be dismissed and the problem confronting the experimental psychologist is to determine those factors which produce the behavior which has been designated "goal orientation."

It is of interest to note that the problem of goal orientation has concerned the "response-reproduction" theorist. Spence (30), Spence and Shipley (31), and Hull (20) have attempted to explain this phenomenon with the use of such concepts as the anticipatory goal reaction and the habit family hierarchy. Although theorists have been interested primarily in problems of response-reproduction, they have been cognizant of the problem of predicting new reponses and have attempted to introduce theoretical formulations to explain such behavior.

In 1946, Tolman and his co-workers published the first of the series of articles referred to in the introduction. In these articles the concept of spatial learning has been defined more specifically and experiments

have been devised to illustrate the difference between place-learning and response learning.

The first experiment (36) in the series of articles published by Tolman and his co-workers was designed to test the hypothesis that an animal which has learned to take a certain path to food will, when that path is blocked, take an alternate path leading to the place at which food was originally found.

An apparatus was used on which the rat learned first to run a path to a goal box located 6 inches in front of a 5 watt lamp. After this training series the original path was blocked and 18 paths radiating from the circular choice point were substituted. Of these 18 possible choices, 36% of the rats on the first test trial took path 6 which was the path which led most directly to the original location of the goal box. The remaining rats were distributed in a "random fashion" over the other paths. The conclusion drawn was that rats "learn to expect goals in specific locations."

One criticism which can be made of the experimental technique of Tolman and his co-workers is that the light was in the immediate vicinity of the correct path (path 6). It might be said that the animals were making a response to the light rather than to the location of the goal. In answer to this criticism Tolman, Ritchie, and Kalish (36,p.20) point out that path 5 which was almost as

brightly lighted as path 6 but which did not point as directly toward the goal was entered by only 4% of the animals. However, the possibility that the response could have been to the light remains. A situation should have been set-up in which there could be no question of the purely spatial character of the correct response.

In order to test this conclusion an experiment was designed to present direct evidence for the hypothesis that in most cases animals do not learn to make a specific response but rather to orient toward the place of the goal. (37) Such a hypothesis had been strongly suggested by the results of the above mentioned experiment.

A T-maze was used which varied from the typical T maze in that the stem of the T could be shifted to either side of the choice point, so that from one starting point the animal would have to turn left to get to the same physical goal box that was reached by a right turn from the other starting point. One group of animals ("response-group") had to make the same spatial response, i.e., to the right, to receive a reward no matter from which side of the maze they were started. The "place-learning group" was to go always to the same location to obtain reward no matter on which side of the maze the stem of the T was placed (i.e. from one starting point the correct turn was to the right and from the other the correct turn was to the left; in both cases the physical location of the food

remained the same.)

The results showed that of 8 rats in the response-learning group only 3 reached the criterion of 10 successive errorless runs. The others in this group were run to 72 trials and never learned. Rather, they formed the habit of always going to the same place. The 8 rats in the place-learning group all reached the criterion within 8 trials. Tolman presents these results as evidence that:

> ...both kinds of dispositions may be acquired by the rat, but that the disposition to orient towards the goal is simpler and more primitive than the disposition to make right turns. (37,p.228)

It is interesting to note that in this experiment Tolman modifies his position to some extent, since he suggests that in some cases rats do learn to make specific responses. In the experiment, however, the inference is that learning a response and learning a place are two different types of phenomenon which are to some extent mutually exclusive.

The third experiment of the series, which was conducted by Ritchie (28) was designed to test a question which arose from the preceding two studies. If one group of rats is trained to take two paths from two separate parts of the room to the same place and another group is trained to take two paths from two parts of the room to two different places, which will learn more rapidly? Ritchie points out that according to the previous experiments the group which learns two paths to one place

should learn the problem more rapidly since this group would be rewarded in only one place. According to the place-learning theory, the training from the different starting places should only emphasize the place of the reward. However, in the group trained to go to two different places from two different locations the acquisition of an adequate cognitive map would be somewhat more difficult.

The initial experiment which was designed to test this hypothesis produced results not in agreement with the prediction. There was no significant difference in the rate of learning between the two groups. This result is explained by Ritchie as indicating that both groups were learning to go to a certain place. The identity of location was planned in one group; but the second group, which was supposedly learning to take two paths to two different places, was in reality, he stated, also learning two paths to the same goal as was the first group. The orientation point for the second group was found to be the cage from which the animals were taken before each trial and to which they were returned after the trial. Both correct choices for this group pointed in the direction of this cage. The stimuli from the cage (noise, odor, etc.) were diffused, thus generalizing the location of the cage so that from both paths it was in the same general place.

When these conditions were changed and the animals run on two different paths to two different places in two different rooms there was a significant difference between the two groups. The group which was trained to take two pathways to the same goal learned more rapidly than the group which was trained to take two different paths to two different goals.

As an explanation of this finding, Ritchie suggests that the expectation acquired by the rat of food in a certain place produced a disposition for the animal to go directly to the place of food from other parts of the room.

The fourth experiment (38) was designed to test an assumption which was implicit in the hypothesis put forward to account for the results in the third experiment. This assumption was stated by Tolman, Ritchie, and Kalish as follows:

> It was assumed that if rats were trained from S_1 [the starting point] to L [the location of food] and then put on S_2 a place they had never been before, they would perform from S_2 better than rats which had not had such preliminary training. (38,p.39)

It was this assumption which the fourth experiment was designed to investigate. The experiment consisted of training the experimental group on a T-maze, similar to the one used in the second experiment, which was constructed so that the stem could be attached to either side of the choice point. The experimental group of rats was

run from one side of the T-maze for five trials a day. Then the stem was switched and the sixth trial was run from the other side of the arms. The food remained in the same place for all trials of each rat. A light was located in one corner of the room in such a position that one arm of the maze was well lighted and the other in a shadow. This provided a cue by which the animals could orient to the location of food.

A control group was given five trials a day on a straight path in another room and then given one trial from the side of the T-maze from which the first group received the sixth daily trial.

The results were inconsistent with the assumption since both groups performed on the sixth trial with the same efficiency. As a possible explanation of these negative findings, Tolman and his co-workers suggested that in most cases the rats were learning both to go to a place and to make a certain turn either to the left or right side. In the sixth trial these two tendencies inhibited each other, making the performance of the group with training on the T-maze approximately the same as that of the group without previous training.

To investigate whetheror not this turn was the inhibiting factor, a new experiment was performed. During the first five daily trials of the first five days of the experiment neither of the stems was attached to the arms of

the T-maze. The rats were started from one end of the arms of the T-maze and allowed to run to the other end, either toward the light or away from the light, to receive food. For the sixth trial on each of the five days, the rats were run from the stem of the T. Food was always placed in the same location. The first four days the stem was placed on one side of the arms in the sixth trial and on the fifth day it was placed on the opposite side for the sixth trial. On the sixth day, five trials were given from one side and the sixth from the other side.

The results indicated that these groups made significantly fewer errors than did the experimental group in the first experiment. The authors conclude, therefore, that, for the experimental set-up used, the following hypothesis is true:

> If rats are trained (in the given room and under the given lighting conditions, etc.) from either the east or west end of the central path of a T-maze to food at the opposite end and then are put on a starting leg on the south side of the arms of the T, they will perform from this starting place better than a group which has not had this preceding training on the central path. (38,p.46)

In this experiment, the assertion is again made that rats can learn both to make a response and to go to the same place. However, no answer is provided to the question of whether learning a place and learning a response are different aspects of the same process or are two different processes.

The recent work of Tolman and his co-workers has been cited in some detail to show the extent and type of work which has been done on spatial learning. Tolman's interest has been primarily focused on discovering whether the animal will go to the location of the food over a new path when the original one is blocked. Up to the present time, Tolman has not published any work designed to discover the specific variables which influence the degree of spatial learning, except in the fourth experiment in which the authors suggest that a greater degree of spatial learning is obtained when the orientation training is acquired on a path which does not involve the problem of having to choose between two possible choices to reach food.

The purpose of the present experiment is to investigate one of the variables which may affect the spatial phenomenon which Tolman has described. This study is not designed to serve as a crucial test between Tolman's place-learning theory and the opposing response-reproduction theories. However, experiments such as this one which investigate the variables determining the degree of spatial learning may make it possible to determine whether the same relationships existing in response-reproduction experiments also exist in spatial learning experiments. If they do, then many of the principles of simple type learning situations, such as classical and instrumental

conditioning, may be applicable to spatial learning. Such experiments may thus reconcile the findings of spatial learning with the findings of the response-reproduction theorists.

The specific problem investigated in this experiment is the determination of the influence of differing degrees of motivation on the degree of spatial learning.

In recent years most theorists have recognized that motivation is one of the most important variables in the psychology of learning. This is true for Hull who has conceived of drive as a factor which combines in a multiplicative way with the inferred concept of habit strength to determine the behavior potential. However, drive is not a factor in determining the habit strength or amount actually learned by the animal. Habit strength, in addition to other variables, is a function of the number of reinforcements which the animal has received. Kendler (23) investigated the adequacy of Hull's formulation of this relationship. By varying the amount of drive during acquisition and extinction trials in the Skinner box, he found that the degree of motivation during the acquisition had no effect upon the number of trials to reach extinction. These results were consistent with Hull's contention that drive is not one of the variables which determine the "strength of a habit." If this relationship also held in spatial learning, it would be found that the amount of

motivation in the initial training trials would not produce differential amounts of spatial learning, whereas differences in motivation in the test trials should produce differences in performance.

In the four experiments by Tolman and his co-workers all the animals were motivated by approximately 22 hours of food deprivation. Consequently, there is no evidence relative to the important problem of how varying degrees of motivation affect the amount of spatial learning.

The present experiment was designed to investigate whether different degrees of motivation during either the training trials or the test trials of a spatial learning problem have any effect upon the amount of spatial learning exhibited. Or, in terms of Tolman's formulation, whether different amounts of motivation affect the "clarity" of the "cognitive map".

One possible hypothesis within the framework of Tolman's theory would be that the stronger the motivation the more limited the "perceptual field" would be, resulting in less knowledge as to those parts of the "cognitive map" not specifically traversed, even though the features of the limited field might be better learned. The deduction from such an assumption would be that the amount of spatial learning would be inversely related to the strength of motivation during the actual learning. However, as Tolman's theory has been formulated and reported, no such

assumption has been made.

EXPERIMENT I

SUBJECTS

The subjects for this experiment were 46 naive, male albino rats of the Wistar strain, purchased from the General Biological laboratory in Chicago, Illinois. At the beginning of the experiment they were approximately 70 days old.

APPARATUS

The floor plans and dimensions of the maze used in this experiment are shown in Figure 1.

Walls and floors were constructed of wood 5½ by 3/4 inches so that the width of all maze paths was four inches. The entire maze was painted a flat black and covered with wire mesh. The doors, designated in Figure 1 by heavy lines drawn at right angles to the maze alleys, were made of plywood with sponge rubber attached to the bottom of each, making them practically noiseless when manipulated in a vertical direction by strings controlled by the experimenter from behind a screen 59 by 19 inches. Black curtains, indicated by wavy lines in Figure 1, were hung six inches from the end of each alley to prevent the animal from observing the end of the pathway.

The length of all alleys except the two diagonals to

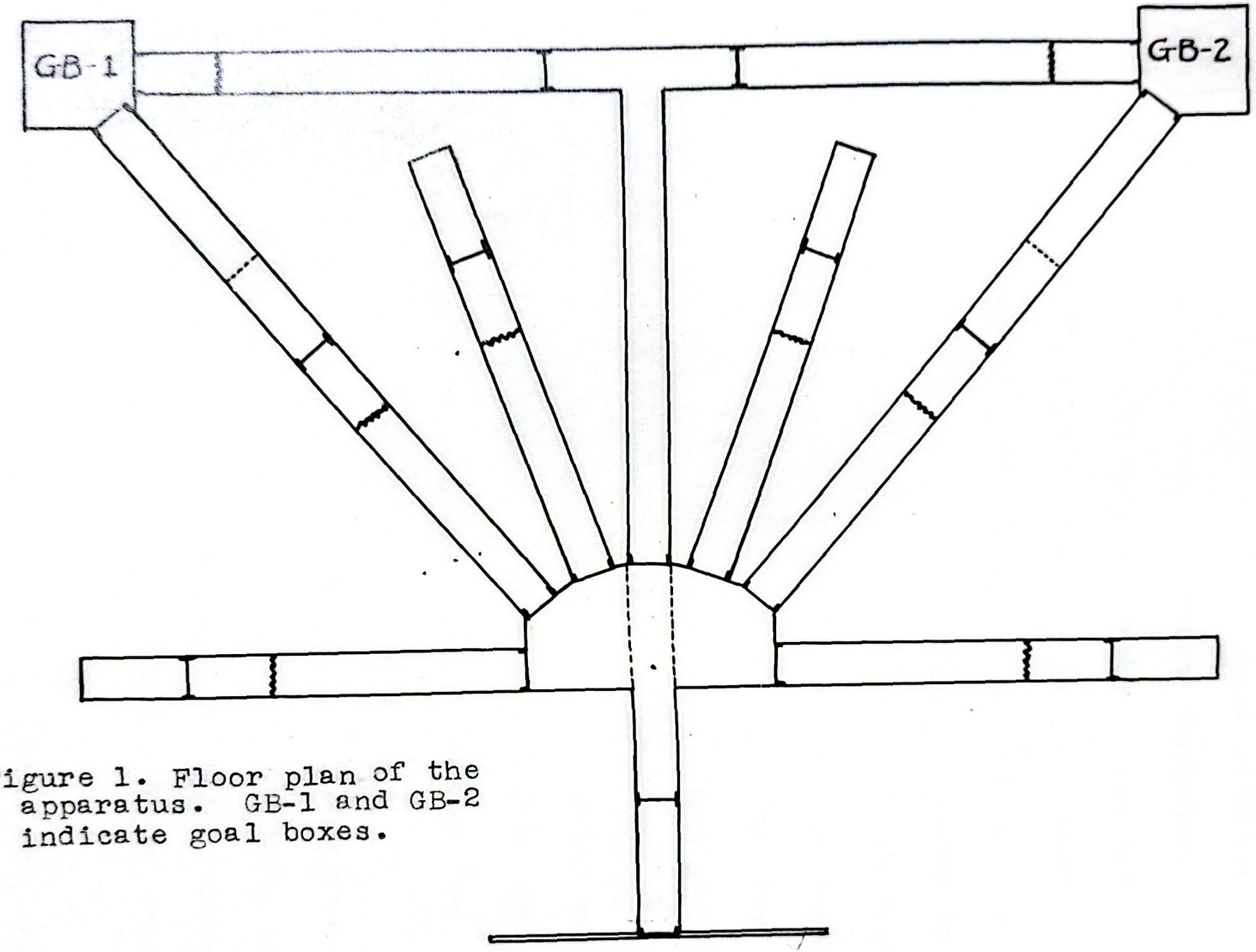

Figure 1. Floor plan of the apparatus. GB-1 and GB-2 indicate goal boxes.

the goal boxes (GB1 and GB2 in Figure 1) was 42 inches. The length of the two diagonals was 62 inches.

The construction of this apparatus permitted the animals to be subjected to two separate maze problems. During the training trials, the removable partition, shown by dotted lines in the semi-circular choice point in Figure 1, was put in place, making a typical T-maze problem. The addition of this removable partition plus the distance to the starting box door added 21 inches to the length of the stem of the T-maze, making a total distance of 63 inches to the choice point. During the test trials this partition was removed, the door to the stem of the T-maze was closed, and the animal was permitted to choose one of the six radiating alleys. The angles of these alleys at the point of intersection, indicated by the dot in Figure 1, were, from left to the center, 90 degrees, 62 degrees and 31 degrees. The angles on the right side were the same as those on the left.

A black wooden block, indicated by the dotted lines in the two long diagonal alleys in Figure 1, was used to make all alleys, except the correct one, the same length. It was inserted during the test trials in the alley which led to the non-rewarding goal box.

The apparatus was in a room 13 by 18 feet. The maze was placed in a corner of the room; a table was placed directly in front of the entrance to the maze, but was

shielded from it by a screen behind which the experimenter stood while the rat was in the maze. The rats were carried from the neighboring room in which the living cages were located in a carrying cage with individual compartments, in which they were kept when not being run.

The only light in the experimental room was that provided by a 150 watt Reflector-Flood bulb placed six feet directly above the choice point of the T maze.

PRELIMINARY TRAINING

From the time of arrival in the animal laboratory the animals were kept in large cages. For the first week after their arrival food was present in the cages at all times. At the end of the week their ears were punched for identification.

The animals were then randomly divided into four groups and each group was placed in a large cage, (11 or 12 rats per cage). At this time, one week before the beginning of the experiment, the animals were placed on a 24 hour wet-food maintainance schedule. Special feeding cages were utilized and the rats were allowed to remain in these cages for 1½ hours during which time they were permitted to consume a wet mash of Purina Kibble meal. The average consumption was approximately 10 grams of food, not including water content, per rat per day. Follow-

ing this 1½ hour feeding period, the animals were returned to their home cage.

Throughout the experiment two different motivational states were utilized. The rats on six hour motivation were given their daily ration 7½ hours prior to their daily experimental trials, resulting in a deprivation period of 6 hours since their feeding time was limited to 1½ hours. The 21 hour group was fed 15 minutes after the completion of the day's runs and the food removed 1½ hours later, resulting in a period of deprivation of food for 21 hours. Water was always available in the home cages.

The position habit of each animal was determined on the basis of three non-reinforced trials on the T-maze. In most cases all three trials were given on the same day. However, in the case of several animals who made no response in 10 minutes the first day, the position habit trials were completed on succeeding days.

All preliminary and experimental trials were run between the hours of 5 p.m. and 1 a.m.

TRAINING TRIALS

The four groups, which had been randomly selected, were referred to as M6-6, M6-21, M21-6, and M21-21. Each of the first two groups contained 11 rats, and each of the last two groups contained 12 animals. In the nomenclature

of these groups the M refers to motivation, the first digit to the number of hours of food deprivation existing during the training trials, and the digit following the hyphen refers to the amount of deprivation during the test trials. Thus two groups, the M6-6 and M21-21, were under the same degree of deprivation in both stages of the experiment, while the M6-21 and M21-6 groups had their motivation switched at the start of the test series.[1]

The T-maze section of the apparatus was used during the training series. Half of the animals were rewarded during the experiment on the side consistent with their position habit, while the other half were rewarded on the side opposed to their position habit.

Each rat was given four trials per day, using a non-correction technique, with at least 15 minutes between trials. If the correct alley was chosen, the rat received in the goal box 1/8 teaspoon of a fine mash of Purina Dog checkers which had been dissolved in water. This food was placed on an inverted metal paint-mixing dish (diameter - 1 1/8 inch) in the goal box. After consuming the reward, the rat was returned to the carrying cage. If the incorrect alley was chosen, the animal was confined in the empty goal

1. The first ten rats which were run in the M6-6 and M6-21 groups (5 from each group) were run under five hours of motivation. Several of these rats would not eat the food found in the goal box. Therefore, the motivation was increased to six hours after three days of running. In no case was a trial, in which the animal went to the correct side but did not eat, counted as a reinforcement.

box for 30 seconds and then returned to the carrying cage. The trials were continued until the animal had received 20 reinforcements, i.e. a total of 20 responses to the correct side followed by the consumption of the food.

Rats number 80 (Group M6-6) and 123 (Group M21-21) made no correct responses in five days. In order to produce learning, a technique of forced trials was instituted. The animal was forced to choose the correct side by the lowering of the door on the side of the choice point which led to the empty goal box. Forced trials were alternated with free trials in the following order--Forced, Free, Forced, Free, Forced, Free, Free, Free. This order of presentation was followed until the animal began to make correct free choices. Rat number 80 was forced to the correct side three times and rat number 123 was forced seven times. If the animal ate the food a forced trial to the correct side was counted as a reinforcement.

TEST SERIES

The day after the 20th reinforcement, the animal was placed in the test situation. The partition was removed from the semi-circular choice point and the door in the stem of the T-maze was lowered, thereby forcing the animals to choose one of the six radiating paths from the new choice point. Before the first test trial of each rat, the semi-circular choice compartment and a distance of one foot into each alley were wiped with a damp cloth to eliminate any possible odor which might have led to track-

ing. As in the training trials, each rat was given four trials per day. The reward was the same (1/8th teaspoon of a mash of Purina Dog Checkers and water), and was in the identical goal box in which it had been during the training trials. The light was also in the same position. The only difference in procedure was that the food was placed on an inverted clear glass coaster (diameter 1½ inches). This change was introduced to eliminate the secondary reward value of the original dish.

All animals were run for 10 days, or 40 trials in the test series. The time taken to reach the end of the chosen alley and the number of alleys entered were recorded. In the recording system, alley 2 was always the correct choice. Thus if an animal received food on the left side of the maze, in GB1 the numbers ran from 1-6 from left to right. If, on the other hand, the animal found food in the right goal box GB2 the system was reversed and the numbers ran from 1-6 from right to left. In such numbering alley number 2 always represented the correct choice during the test series and alley number 1 represented the 90 degree angle pathway next to alley number 2.

At the beginning of the experiment it was planned to have each rat reach a criterion of 12 successive correct responses over a period of three days. At the end of 12 days of running during the test trials only a few rats had reached this criterion, while most of the others indi-

cated no appreciable signs of learning. It was then decided to change the period of the test trials to 10 days for all animals. For those rats which had reached the criterion in less than 10 days before the change in procedure, it was assumed that they would have continued to make perfect responses during the remainder of the 10 day period. Six animals reached the criterion in less than ten days. They were: numbers 109 and 113 in Group M6-6 who reached the criterion in 6 and 5 days respectively; number 75 in Group M6-21 who required 8 days to reach the criterion; number 70 in Group M21-6 who reached the criterion in 7 days; numbers 52 and 55 in Group M21-21 who reached the criterion in 8 and 9 days respectively. [2]

RESULTS AND DISCUSSION

The main experimental results used to answer the question of what effect motivation has on spatial learning, were the choice on the initial test trial and the number of correct responses in 20 and 40 trials of the test series.

As has been stated, alley number 2 was always the

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2. The change of criterion not only served the exigencies of the situation but also had some advantages over the one initially planned. By providing a fixed number of trials, the influence of the learning variable of the test series was reduced.

correct alley. The numbers were assigned to the alleys by reference to the goal box in which the individual rat found food in the training series. If the right goal box contained food, the right 90 degree alley was designated alley 1, the 62 degree correct alley, alley 2, and so forth in a counter-clockwise fashion. If the left goal box contained food, the left 90 degree alley was alley 1 and so on in a clockwise order.

The numbers and percentages of the animals in each group entering each of the alleys on the first test trial are given in table 1. It will be recalled that alley 2 represents the path which led directly to the goal box in which the animal found food during the training trials.

TABLE 1

Showing the number and percent of each group responding to each of the six alleys on the first test trial.

Group	N	Alley 1		Alley 2		Alley 3		Alley 4		Alley 5		Alley 6	
		#	%	#	%	#	%	#	%	#	%	#	%
W6-6	11	5	45.45	0	0	1	9.09	0	0	0	0	5	45.45
W6-21	11	4	36.36	1	9.09	1	9.09	1	9.09	0	0	4	36.36
W21-6	12	2	16.67	1	8.33	1	8.33	1	8.33	2	16.67	5	41.76
W21-21	12	5	41.67	1	8.33	1	8.33	2	16.67	0	0	3	25.00
Total	46	16	34.78	3	6.52	4	8.70	4	8.70	2	4.35	17	36.96

These results are at marked variance with those reported by Tolman, Ritchie, and Kalish (36). They found that 36% of the rats made the correct choice on the first trial, while in the present study 6.52% of the combined groups made the appropriate response on the initial test trial.

Chance factors alone would predict that 16.67% of the animals would choose the correct alley (alley number 2) on the first trial. The standard error of this percent ($N = 46$) is 5.50. The actual percent of responses to alley 2 (6.52%) is 1.85 standard error units away from the chance expectancy. Consequently, by chance this finding would occur, in the direction obtained, less than 5% of

the time.[3]

Table 1 shows that alleys 1 and 6 were entered more often than any other alleys. The total percentage of responses to these alleys were respectively, 34.78% and 36.96%. Therefore, the actual percentage of responses to alley 1 is 3.29 standard error units away from what would be expected by pure chance, while the percent of responses to alley 6 is 3.69 standard error units away from chance expectancy. Thus both alley 1 and alley 6 were entered more than would be expected by chance beyond the 1% level of confidence. Therefore, these results suggest that the animals did not respond to all the alleys in a random fashion on their initial trial, but rather had decided preferences for both 90 degree alleys (alleys 1 and 6).

When the entries into the different alleys made by the four groups are compared with each other (see table 9 in the appendix), it is evident that different motivational states in either the training trials or the test trials produced no significant differences in the pattern of responses to the six alternative pathways on the first

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3. The statistical formulas used in deriving these results were:

$$\sigma_{\%} = \sqrt{\frac{x_{\%}(100 - x_{\%})}{N}}$$

test trial, i.e. all groups exhibited essentially similar behavior.

Table 2 indicates the critical ratios obtained by comparing the number of entries into alleys 1 and 2 and the entries into alleys 6 and 2 made by the subjects in each group. [4]

TABLE 2

Showing the critical ratios obtained by comparing the number of entries into alleys 1 and 2 and alleys 6 and 2 made by the groups. [5]

GROUP	ALLEY 1/ALLEY 2		ALLEY 6/ALLEY 2	
	CR	Level of Significance	CR	Level of Significance
M6-6	3.03	1%	3.03	1%
M6-21	1.61	11%	1.61	11%
M21-6	.62	54%	2.04	4%
M21-21	2.04	4%	1.12	26%
Total	3.57	1%	3.81	1%

4. Complete tables giving the critical ratios between all alleys for each group and for the total number of animals can be found in Tables 10,11,12,13 and 14 in the Appendix.

5. The statistical formulas used in deriving these results were:

$$CR \quad \frac{\text{Obs.Diff.}}{\text{diff.}}$$

$$\sigma_{diff} = \sqrt{\sigma_{x\%}^2 + \sigma_{y\%}^2}$$

$$\sigma_{\%} = \sqrt{\frac{X\%(100-X\%)}{N}}$$

The data contained in Table 2 indicates that there is a tendency in all groups for both alleys 1 and 6 to be entered significantly more often than alley 2. When the same comparisons are made between the total number of animals in the experiment (table 2) this difference is significant. The total number of animals entering alley 1 or alley 6 on the first test trial is significantly greater than the number of those entering alley 2 at the 1% level of confidence.

Although the results of the first test trial of this experiment are at variance with those of Tolman, Ritchie, and Kalish, there are several important methodological differences which can probably account for such divergencies. These possible influences will be discussed at a later point in the study.

The data relating most directly to the problem of the relation of motivation to the degree of spatial learning were obtained by comparing the number of correct responses made by the different motivational groups in 20 and 40 trials. Two general comparisons were made:

1. A comparison between those groups with different motivation during the training series and the same motivation during the test series.

2. A comparison between those groups with the same motivation during the training series and different motivations during the test series.

If motivation during the training series had an effect on spatial learning, then we should expect a difference in

the following two comparisons between groups which had different motivation during the training trials and the same motivation during the test trials: (Groups M6-21/M21-21 and Groups M6-6/M21-6). Table 3 indicates the comparisons between the number of correct responses (responses to alley 2) made by these groups in 20 and 40 trials.[6]

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6. It would have been desirable to use a statistical technique such as the analysis of variance which would provide an over-all statistic evaluating the influence of motivation on spatial learning. However, it was not possible to do so since the assumption of approximately equal variances in all groups which is basic to the analysis of variance was not met. In the case of unequal variances a significant F might be due to the difference in variances and not to the difference in means. Therefore, a technique devised by Festinger (9) for determining the significance of means with a skewed distribution was used.

TABLE 3 [7]

Comparisons of differences between mean number of correct responses during 20 and 40 test trials. (x)

	20 Trials				40 Trials			
GROUP	MEAN	F-RATIO	DF	SIG.	MEAN	F-RATIO	DF	SIG.
M6-6	4.18		22		11.64		22	
M21-6	4.83	1.16	31.09	-	16.08	1.38	61.00	-
M6-21	3.36		52.63		14.55		83.69	
M21-21	5.08	1.51	55.06	20%	16.17	1.11	99.25	-

(x) Comparisons are between groups that had the same motivation during the test series and different motivation during the training series.

As shown in table 3, one of the four comparisons between groups with different motivation during the training trials and the same motivation during the test series is significant at the 20% level of confidence. This is obviously not sufficient to reject the null hypothesis and therefore the hypothesis that the "clarity" of the "cognitive map" is independent of motivation, re-

7. The statistical formulas used in deriving these results were:

$$\text{F-Ratio} = \frac{M_1}{M_2}$$

$$\text{Degrees of freedom} = 2NP$$

$$P = \frac{M^2}{V}$$

$$V = \sigma^2$$

mains tenable.

To discover the effect of the same motivation during the training trials and of different motivation during the test series, comparisons between M6-6 and M6-21, and M21-6 and M21-21 were made. If motivation during the test series is an important factor, these comparisons should be significant.

TABLE 4

Comparisons of differences between mean number of correct responses during 20 and 40 test trials. (x)

	20 Trials				40 Trials			
GROUP	MEAN	F-RATIO	DF	SIG.	MEAN	F-RATIO	DF	SIG.
M6-6	4.18	1.24	22	-	11.64	1.25	22	-
M6-21	3.36		52.63		14.55		83.69	
M21-6	4.83	1.05	31.09	-	16.08	1.01	61.00	-
M21-21	5.08		55.06		16.166		99.25	

(x) Comparisons are between groups that had the same motivation during the training series and different motivation during the test series.

There is no significant difference between the mean number of correct responses made by the groups with the same motivation during the training series but different motivation during the test trials. Therefore, it is again impossible to reject the null hypothesis.

The results are consistent in all comparisons between the various motivational groups for both the 20 and 40 trials in indicating no significant difference. Thus,

differing degrees of motivation in either the training or the test series had no significant effect on the degree of spatial learning as measured by the number of correct choices in 20 and 40 trials. These results are relative only to the difference in motivation developed by 6 and 21 hours of food deprivation. If this difference had been greater, for instance 3 and 48 hours of food deprivation, it might have been found that the degree of motivation did effect spatial learning. However, for the motivational states used, motivation had no effect on the degree of spatial learning.

The results show that groups having different degrees of deprivation during the training trials and the same degree of deprivation during the test trials do not differ significantly in the amount of spatial learning exhibited. This would be consistent with Hull's (19) assumption that drive is not a factor in the formation of habit strength (amount of learning that takes place). In this case, relationships developed by response-reproduction theorists to account for data from conditioning situations may be applied to the results of a spatial learning problem.

However, the results revealing no significant difference in the amount of spatial learning exhibited by the groups having different motivational states during the test series, when motivation for the training series is equated, would be inconsistent with the formulations of

Hull (19). The difference in motivation, while not producing a difference in the amount actually learned, should produce a difference in the actual behavior because of the greater "multiplicative" effect of a 21 hour hunger drive as compared with a 6 hour hunger drive. One possible explanation for this inconsistency is that there was so little transfer between the training and test situations that the different drives were not sufficient to reveal a difference in 40 trials. Perhaps, if the experiment had been carried out for a longer period of time the difference might have become apparent.

In addition to the main results discussed above, other data were gathered relative to the performance of the animals in the experimental situation.

The mean number of errors to reach 20 reinforcements in the training series by the animals under 6 hours of deprivation (Groups M6-6 and M6-21) and those under 21 hours of motivation (Groups M21-6 and M21-21) and the significance between these two groups is given in Table 17 in the appendix.

It would be expected that the rats under 21 hour motivation would learn with significantly fewer errors than those under 6 hours of motivation. However, although the mean number of errors made by the 6 hour group (7.42) is more than that made by the 21 hour group (6.89), the difference is not significant. However, when we measure

the strength of response in terms of latency (Table 18 in the appendix) we discover that the mean time of the first correct response on the last day of the training trials for the groups under 6 hours of motivation (Groups M6-6 and M6-21) was significantly more than that of the groups under 21 hours of motivation (Groups M21-6 and M21-21). This indicates that motivation did have an effect on the behavior of the animals.

The average time, for each group, of the first correct response on the last day of the training trials was compared with the average time taken by each group to make the first response in the test series. The mean of each group in both the training and test series and the significance between these two means is shown in Table 19 in the appendix. The difference is significant in all cases and indicates that the new situation of the initial test trial greatly increased the latency of the animals response to the choice point situation.

The mean time of the first response in the test trials was also compared between the four groups to discover if the difference in motivation had any effect on the time taken to make the first test response. The results shown in Table 20 of the appendix, of the comparison of groups having the same motivation during the training series and different motivation during the test series and of groups having different motivation during the training series and

the same motivation during the test trials indicate that there is no significant difference between the latencies of groups having the same motivation during the training trials and different motivation during the test series, (Groups M6-6:M6-21 and Groups M21-6:M21-21). However, when comparisons are made between the groups with different motivation during the training series and the same motivation during the test series (comparisons between M6-6:M21-6 and M6-21:M21-21) it is found that the differences are significant. Combining these significances by the chi-square method of combining probabilities, it is found that the groups which had 21 hours of motivation during the training series made the first choice of the test series more rapidly than those which had 6 hours of motivation during the training series at the 5% level of confidence. Motivation during the training trials seems thus to have more effect than motivation during the test trials in determining the time of the first test trial. The animals under 21 hours of food deprivation probably learned to run more quickly. The speed of a response is one aspect which is learned, according to response-theorists.

The number of responses to alleys 1, 2, and 3 during the test series for the different motivational groups were tabulated and the appropriate comparisons made. Although only one alley (alley 2) led directly to food, alleys 1 and 3 both were on the same side of the stem of the maze.

which contained the goal box with the food reward. Consequently, responses to these alleys provide us with another measure of spatial learning. These results, given in Tables 15 and 16 in the appendix, show that differences in motivation do not produce any significant differences in the number of times the correct side of the maze was chosen in the test series. These results are, therefore in complete agreement with the other criterion of spatial learning analyzed.

The results of this experiment differ most widely from those obtained by Tolman and his co-workers in the number of rats choosing the correct alley on the first test trial. It will be remembered that in the present experiment 6.52% of the animals chose the correct alley on the first spatial trial, whereas in the first experiment by Tolman and his co-workers 36% of the animals chose the correct path from 18 possible choices.

Many differences in apparatus and technique could be responsible for this discrepancy. The most apparent dissimilarity between the two pieces of apparatus is that in the present experimental set up there were fewer cues by which the animal could orient towards the location of the goal. The maze used in this experiment was not an elevated maze such as used by Tolman, Ritchie, and Kalish. The light in the present study was directly above the

choice point of the T-maze, making the lights and shadows almost identical in opposite parts of the maze. It was possible that going to the left or right of the light did not provide a strong enough cue for the animal. Therefore, it was decided to repeat the experiment, changing the position of the light to provide a stronger orienting cue. This was done to discover if such a situation would produce more spatial learning. Accordingly, a light was placed directly over the right goal box (GB 2) in Experiment II, at a height of 4 feet.

EXPERIMENT II

SUBJECTS

Twenty naive, white rats, eleven males and nine females, which had been bred at the University of Colorado laboratory were used as subjects. They were litter mates from two litters and were approximately 60 days old at the start of the experiment.

APPARATUS

The apparatus used was the same as that used in the first experiment. The experimental room and all conditions in the room were identical except that a light of the same intensity and make as that used in Experiment I was hung directly above the right goal box, 4 feet from the floor. This light made the right arm of the T-maze and alley 2 very light while the other alleys were in varying degrees of shadow. The shadow was deepest in alleys 6 and 5. The light used in Experiment I was turned off during the running of this experiment.

PRELIMINARY TRAINING

The animals were randomly divided into two groups,

LM6-21 containing six males and four females, and LM21-21 containing five males and five females. The nomenclature is similar to that utilized in the initial experiment. The L preceding the M indicates the change in the position of the light. The food and feeding schedule remained the same in this experiment.

During the experiment, the animals were always run between the hours of 3:00 p.m. and 8:00 p.m. The experimental session involving the female subjects always followed the one involving males. After the females had run, the maze was wiped out with a damp cloth to prevent any tracking.

TRAINING SERIES

Half the animals of each group found food on the side consistent with their spatial preference whereas the other half were run against their position habit. All animals received food in the right, lighted goal box. As in the initial experiment, each animal was required to make 20 correct responses in the T-maze situation and to consume the food following the appropriate response. [8]

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8. The experimental procedure was varied in some cases in this experiment. Four animals in the LM6-21 group would not run when placed in the maze for some unknown reason. To force these animals to run, they were put in the lighted goal box and allowed to eat. Then they

TEST SERIES

As in Experiment 1, the T-maze was closed and the animals forced to make a choice from the semi-circular choice point. The only deviation from the procedure of the first experiment was that the animals were run for only 20 trials.

RESULTS AND DISCUSSION

To discover the effect of motivation on spatial learning in Experiment 2, the most important results studied were:

1. The first choice of each animal on the test trials.
2. The number of correct choices (alley 2) made in 20 test trials. (In Experiment 2 the animals were only run for 20 test trials.)

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were put into the right arm of the T-maze with the door closed and allowed to run to the goal box and eat. After this training, they were put into the start of the maze and run in the usual way to 20 reinforcements. One animal in the LM21-21 group had made no correct responses after five days of running. When an attempt was made to force a correct response in the same manner used in Experiment 1, the animal would not run in the lighted alley and found a way to open the door into the main portion of the maze. The same technique of feeding and running as was described above was used with this animal.

The results obtained from the five rats treated in the above manner were so divergent from the results of the other animals in the amount of spatial learning shown that they have been excluded from the main results and will be discussed separately.

The number and percentage of the rats in each group entering each of the alleys on the first trial is given in table 5.

TABLE 5

Showing the number and percent of each group responding to each of the six alleys on the first test trial.

		ALLEY 1		ALLEY 2		ALLEY 3		ALLEY 4		ALLEY 5		ALLEY 6	
GROUP	N	#	%	#	%	#	%	#	%	#	%	#	%
LM6-21	6	1	16.67	0	0	0	0	0	0	2	33.33	3	50.
LM21-21	9	1	11.11	1	11.11	1	11.11	1	11.11	1	11.11	4	44.44
Total	15	2	13.33	1	6.67	1	6.67	1	6.67	3	20	7	46.67

As in experiment 1, chance factors alone would predict that 16.67% of the animals would choose each alley on the first test trial. The standard error of this percent is 9.66. The only significant variation from chance expectancy occurred in the percent of responses to alley 6 on the first test trial which is 3.10 standard error units away from the chance expectancy. Thus, in less than 1% of the cases would this difference be attributable to chance.

As was found in Experiment 1, there is no significant difference between the groups in the percent of choices made to each alley on the first test trial. (See Table 21 in the appendix) This again suggests that the motivation

of the animals during either the training or the test series had no effect on the choice made on the first test trial.

Table 6 indicates the critical ratios of the number of entries into alleys 1 and 2 and alleys 6 and 2 by each group and by the total numbers of rats in the experiment. [9]

TABLE 6

Showing the critical ratios obtained by comparing the number of entries into alleys 1 and 2 and alleys 6 and 2 made by the groups.

GROUP	ALLEY 1/ALLEY 2		ALLEY 6/ALLEY 2	
	CR	SIG	CR	SIG
IW6-21	1.10	27%	2.45	2%
IW21-21	0	0	1.70	9%
TOTAL	.61	54%	2.78	1%

The choices made on the first test trial differ in Experiment 2 from those first choices made in Experiment 1 in the percent choosing certain alleys. It will be remembered that in Experiment 1 alleys 1 and 6 were both chosen significantly more than alley 2 on the first test trial. In Experiment 2, however, there was no significant differ-

9. For complete tables giving the critical ratios between entries into all alleys for both groups and the total number of animals see Tables 22,23, and 24 in the appendix.

ence between the entries into alley 1 on the first trial and into alley 2. However, alley 6 was entered significantly more than alley 2 on the first test trial at the 1% level of confidence.

Why, in Experiment 2 which was designed to provide more cues to the location of the goal, should a greater percent of animals choose alley 6 (the alley farthest from the lighted goal box) on the first test trial? The most plausible hypothesis would seem to be that the initial test trial produced a marked emotional response in the rat. This is borne out by the observation that in the first test trial there was more urination and defecation than had been observed since the first few trials on the T-maze. (Both urination and defecation are recongized as emotional responses of the rat (14)). Another emotional response of the rat is that of going away from the light toward a dark location. In Experiment 1, alleys 1 and 6 were the darkest alleys. However, in Experiment 2, alleys 6 and 5 were the darkest (it will be noted, Table 6, that a greater percent of animals entered alley 5 on the first test trial than did in Experiment 1.) Thus, it is suggested that one of the main causes of the lack of spatial responses on the first test trial was the emotional condition of the rats.

It has also been suggested previously in this study that the construction of the maze used in the present experi-

ment may have made spatial learning more difficult than the elevated mazes used by Tolman and his co-workers. Further experiments would be necessary to determine the effect of this type of apparatus on spatial learning. It is possible that if a dim light had been used there would be a reduction in the emotional response. Another possible experiment to reduce the emotional response would be one in which the animals were put through the training series without having the partition in place which changed the semi-circular choice point into a straight alley. This would allow the animal to become familiar with the choice point used in the test series before the test trials are run and would thus eliminate emotional reactions resulting from a completely new situation.

The primary measure of the relation of motivation to spatial learning in Experiment 2 was the comparison of the number of correct responses made by groups having different motivation during training trials and the same motivation during the 20 test trials. These data are presented in Table 7. No data were obtained in Experiment 2 with regard to the effect of the same motivation during the training trials and different motivation during the test trials.

TABLE 7

Comparison of differences between mean number of correct responses during 20 test trials. (x)

GROUPS	MEAN	F-RATIO	DF	LEVEL OF SIGNIFICANCE
LM6-21	6.83	1.12	34.72	-----
LM21-21	7.67		108.22	

(x) Comparisons are between groups that had different motivation during the training trials and the same motivation during the test series.

The fact that this comparison shows that the difference is again not significant supports the results of Experiment 1 and indicates that we cannot reject the null hypothesis. This suggests that for the degree of food deprivation studied, the amount of motivation does not affect the degree of spatial learning. Or, in Tolman's terms, the degree of motivation during the training series does not change the clarity of the "cognitive map" learned by the rat, and thus does not affect the spatial learning displayed.

To determine whether or not the degree of spatial learning in the second experiment was greater than that demonstrated in the first experiment, a comparison was made of the comparable groups in the two experiments. This comparison is presented in table 8. [10]

10. The validity of this comparison is somewhat lessened by the fact that it was necessary to use two different

TABLE 8

Comparison of differences between the mean number of correct responses during 20 test trials between comparable groups of Experiments 1 and 2.

GROUPS	MEAN	F-VALUE	DF	SIG
W6-21	3.36	2.03	52.63	5%
LW6-21	6.83		34.72	
W21-21	5.08	1.51	55.06	5%
LW21-21	7.67		108.22	

Both of the groups in Experiment 2 made more correct responses than the comparable group in Experiment 1. These differences between the mean number of correct responses are both significant at the 5% level of confidence. When the two significances of 5% each were combined it was found that this combined figure was significant at the 2% level of confidence. This indicates that the change in the position of the light which was the major difference between the two experiments resulted in an increase in the amount of spatial learning exhibited in the test series. The above result serves to emphasize the assertion of Tolman, Ritchie, and Kalish in their first article (36) that a strong orienting cue is necessary for

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stocks of albino rats for Experiments 1 and 2. It is impossible from the present data to evaluate the significance of this experimental variation.

spatial learning to take place.

As in Experiment 1, additional data were gathered on the performance of the animals in the experimental situation.

The significance between the mean number of errors to reach 20 reinforcements in the training period is given in Table 27 in the appendix. In this experiment, the group under 21 hour motivation during the training period (LM21-21) made significantly fewer errors than the group under 6 hours of motivation (LM6-21). This is the result that was expected but did not occur in Experiment 1 on the basis of other experiments testing the rate of learning under different degrees of motivation. The factor accounting for this difference between Experiment 1 and Experiment 2 is not known.

As in the first experiment, the 6 hour group (LM6-21) took a significantly longer time to make the first correct trial run on the last day of the training trials than did the 21 hour group (LM21-21). These results are given in Table 29 of the appendix. This again shows that the factor of motivation influences the rate of behavior but not the amount actually learned.

Table 30 in the appendix gives the means and significant differences found between the first correct choice on the last day of the training trials and the first choice in the test series. These results are similar to those

obtained in the first experiment and show, by the longer time taken to make a choice, the effect the new situation has on the rate of response. Tolman also found (36) that the time taken by the rat to choose one of the 18 paths was quite long.

The result found in Experiment 1 that animals under 21 hour motivation during the training series made the first choice on the test trials in a shorter time than did the animals under 6 hours motivation in the training trials is upheld by the data of Experiment 2. Table 31 in the appendix illustrates this by indicating that the LM21-21 group ran the first test trial at a rate that was significantly faster than the rate at which the LM6-21 group ran the first test trial at the 5% level of confidence. This is the only instance in which the amount of motivation during the training period seems to have any effect on the performance of the animal in the test series and indicates that the habit of a faster response is carried over to at least the first response in a different situation.

Table 25 in the appendix shows that, as in Experiment 1, neither group made significantly more responses to the correct side (alleys 1, 2, and 3) than did the other. When the comparable groups in the two experiments are compared (Table 26 in the appendix) it can be seen that whereas the groups in Experiment 2 made more responses to

the correct side, possibly because of the orienting factor of the light, the difference between the groups in the two experiments is not significant.

The data of the four animals in the LM6-21 group which were given training in running down the lighted alley of the T-maze, (as described in the procedure for Experiment 2), were not included in the results given above because of its deviation from the other data recorded. The mean number of responses to the correct alley made by these four rats in 20 test trials was 12.5. When this is compared with the mean of the balance of the group (6.83), it is found that this difference is significant at the 5% level of confidence. (See Table 32 in the appendix) As this procedure was not well controlled, (i.e. no definite procedure was followed), no conclusions can be drawn from this data. It should be noted that in the fourth experiment of Tolman, Ritchie, and Kalish (38) a technique similar to that used on the four animals in the LM6-21 group was used. The authors also found that this technique increased the degree of spatial learning.

CONCLUSIONS

Three conclusions can be drawn from these data.

1. The first test trial in both experiments gave no indication of spatial learning. The animals did not enter the correct alley as would have been predicted from the work of Tolman, Ritchie and Kalish, but rather tended to choose the two 90 degree alleys which were parallel to the alleys of the T-maze. In Experiment 1 the two 90 degree alleys (alley 1 and 6) were chosen with almost equal frequency and significantly more than any other alley, whereas in Experiment 2 the 90 degree alley farthest from the lighted goal box (alley 6) was entered more frequently than any other alley. It has been suggested that this behavior which is inconsistent with that reported by Tolman, Ritchie, and Kalish (36) can be attributed in part at least to the emotional factor produced by the new situation. The presence of an emotional factor was indicated by the fact that in both experiments the alleys most frequently chosen were those in the deepest shadow. This behavior was most apparent in Experiment 2 in which alley 6 was in deep shadow; in Experiment 1 alleys 1 and 6 were the least illuminated pathways of all the possible choices. The fact that the greatest percent of the rats chose the darkest alleys in both experiments and did not enter the lighted alleys suggests that it was the relative

darkness of the alleys which determined the choice of the animals rather than the position of the alleys. This assertion is emphasized by the fact that when the illumination of alley 1 was increased in Experiment 2 it was no longer entered significantly more than the other alleys. Therefore, definite statements concerning the effect of differing degrees of motivation on the first spatial trial cannot be made as the findings might have been very different had the emotional factor not been present.

2. The results comparing the amount of spatial learning for the various motivational groups for both 20 and 40 test trials provide no significant evidence to support the contention that different degrees of motivation either during the training or test trials affect the amount of spatial learning. It should be realized, however, only two points (6 and 21 hours of food deprivation) on the dimension of hunger were measured. If this difference had been greater a difference between groups might have been found.

Although the animals in Experiment 2 made significantly more correct responses in the test situation than did the animals in Experiment 1, in neither experiment was there any evidence that different motivational states during the training period produced different degrees of spatial learning. The results of the two experiments are in agreement and it can be assumed that, in Tolman's terms, the clarity of "the cognitive map" is not affected by the

degree of motivation during the training series. It should be noted that these results are in agreement with Hull's assumption that motivation affects the performance following learning but not the amount actually learned in a situation in which a definite number of reinforcements are received. Thus, in this case an assumption of a response-reproduction theorist is consistent with the results obtained in a spatial learning problem.

3. The fact that more spatial learning was found in the 20 test trials of the animals in Experiment 2 than was shown in the same number of trials in Experiment 1 indicates that the change in the position of the light was a factor producing this greater spatial learning. This result emphasized the assertion of Tolman and his co-workers that if spatial learning is to occur an orientating factor is necessary. It appears that the more distinct the cue the more spatial learning will be observed. It should be stated that this result may possibly constitute a criticism of the spatial learning theory. Perhaps the factor which determined the increase in the number of correct responses made during the test series was not that the location of the goal was more clearly marked but that the animals made the correct response of going directly toward the light. As has been indicated in the discussion of the first article of the present series by Tolman, Ritchie, and Kalish (36) an experiment

should be done the results of which would not have these two possible interpretations.

SUMMARY

The purpose of this study was to investigate the relationship of motivation to the degree of spatial learning. 46 animals were divided into four groups having either 6 or 21 hours of food deprivation. In two groups, the motivation was changed between the training trials and the test trials, and in the other two groups the motivation was held constant during the experiment.

An apparatus was employed which made possible both the use of a T-maze during the training series and a maze with six choices in the test series, one of the choices leading to the T-maze goal box. A light was placed directly over the choice point of the T-maze. In the training series the animals received 20 reinforcements on the T-maze following the correct choice. After this training, 40 test trials were given on the six-choice portion of the maze.

On the first test trial the animals tended to enter the two 90 degree angle alleys which were parallel with the arms of the T-maze. There was no significant differences between the different motivational groups. It was also found that motivation did not affect the degree of spatial learning as measured by the number of correct responses in 20 and 40 trials.

A second experiment was run in which the conditions

were changed to provide a more distinct cue by which the animals could orient toward the goal. This was accomplished by putting a light directly over the correct goal box. The same amounts of deprivation were used; however only two groups were run: one group having different motivation during the training and the test series and the other having the same motivation during both parts of the experiment.

The results agreed with those of Experiment 1 in that the different motivational states did not affect the amount of spatial learning shown in 20 trials. However, the animals in Experiment 2 gave more evidence of spatial learning in the 20 test trials than did those in Experiment 1 in a similar number of test trials. This indicates the importance of the change in the position of the light.

On the first test trial of Experiment 2, it was found that the animals tended to take the darkest alley. This was attributed to an emotional factor which was probably also in operation during Experiment 1.

APPENDIX

TABLE 9

Indicating critical ratios of the differences between entries of the four experiment groups into the six alleys for the first test trial.

Group	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
M6-6 / M6-21	.44	1.05	0	1.05	0	.44
M6-6 / M21-6	1.56	1.04	.07	1.04	1.55	.18
M6-21 / M21-21	.26	.07	.07	.55	0	.59
M21-6 / M21-21	1.40	0	0	.62	1.55	.88

TABLE 10

Indicating the critical ratios obtained by comparing the percent of entries into each alley with the percent of entries into each of the other alleys for the animals in Group M6-6 on the first test trial. (Experiment 1.)

	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
Alley 1	--	3.03	2.10	3.03	3.03	0
Alley 2	--	--	1.05	0	0	3.03
Alley 3	--	--	--	1.05	1.05	2.10
Alley 4	--	--	--	--	0	3.03
Alley 5	--	--	--	--	--	3.03

TABLE 11

Indicating the critical ratios obtained by comparing the percent of entries into each alley with the percent of entries into each of the other alleys for the animals in Group M6-21 on the first test trial. (Experiment 1.)

	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
Alley 1	--	1.61	1.61	1.61	2.51	0
Alley 2	--	--	0	0	1.05	1.61
Alley 3	--	--	--	0	1.05	1.61
Alley 4	--	--	--	--	1.05	1.61
Alley 5	--	--	--	--	--	2.51

TABLE 12

Indicating the critical ratios obtained by comparing the percent of entries into each alley with the percent of entries into each of the other alleys for the animals in Group N21-6 on the first test trial. (Experiment 1.)

	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
Alley 1	--	.62	.62	.62	0	1.40
Alley 2	--	--	0	0	.62	2.04
Alley 3	--	--	--	0	.62	2.04
Alley 4	--	--	--	--	.62	2.04
Alley 5	--	--	--	--	--	1.40

TABLE 13

Indicating the critical ratios obtained by comparing the percent of entries into each alley with the percent of entries into each of the other alleys for the animals in Group M21-21 on the first test trial. (Experiment 1.)

	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
Alley 1	--	2.04	2.04	1.40	2.93	.88
Alley 2	--	--	0	.62	1.04	1.12
Alley 3	--	--	--	.62	1.04	1.13
Alley 4	--	--	--	--	1.55	.51
Alley 5	--	--	--	--	--	2.

TABLE 14

Indicating the critical ratios obtained by comparing the percent of entries into each alley with the percent of entries into each of the other alleys for all the animals in Experiment 1 on the first test trial.

	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
Alley 1	--	3.57	3.20	3.20	3.98	.22
Alley 2	--	--	.40	.40	.46	3.81
Alley 3	--	--	--	0	.85	3.43
Alley 4	--	--	--	--	.85	3.43
Alley 5	--	--	--	--	--	4.22

TABLE 15

Showing the significance between the number of responses to the correct side (responses to alleys 1, 2, 3) made by the four experimental groups in 20 trials. (Experiment 1.)

Group	Mean	F Ratio	DF	Level of Significance
M 6- 6	13.182	1.1885	120.4126	
M 6-21	11.0909		91.6212	
M 6- 6	13.182	1.2075	120.4126	
M 21- 6	10.917		154.0080	
M 6-21	11.0909	1.2021	91.6212	
M 21-21	13.333		380.2608	
M 21- 6	10.917	1.2213	154.0080	
M 21-21	13.333		380.2608	

TABLE 16

Showing the significance between the number of responses to the correct side (responses to alleys 1, 2, and 3) made by the four experimental groups in 40 trials. (Experiment 1.)

Group	Mean	F. Ratio	DF	Level of Significance
M 6- 6	25.545	1.0321	97.9880	
M 6-21	26.364		154.5214	
M 6- 6	25.545	1.0321	97.9880	
M 21- 6	26.25		171.6288	
M 6-21	26.364	1.1727	154.5214	
M 21-21	30.917		1079.9208	
M 21- 6	26.25	1.1778	171.6288	
M 21-21	30.917		1079.9208	

TABLE 17

Comparing the number of errors made by animals motivated by 6 hours of food deprivation and those motivated by 21 hours of food deprivation to reach 20 reinforcements.

Number of hours of Food Deprivation	No.	Mean	F Ratio	DF	Level of Significance
6 hr.	22	7.418		47.7890	
			1.0766		
21 hr.	24	6.891		73.598	

TABLE 18

Comparing the time in seconds taken to make the first correct response on the last day of the training trials by the animals motivated by 6 hours of food deprivation and those motivated by 21 hours of food deprivation. (Experiment 1)

Motivation	Mean Time	F Ratio	DF	Level of Significance
6 hour	33.409	3.2594	44	1%
21 hour	10.25		48	

TABLE 19

Comparing the mean time in seconds taken on the first correct choice on the last day of the training trials with the mean time taken on the first test trial for the four experimental groups.
(Experiment 1)

Group	Time of 1st Correct Trial on last day of Training Trials		Time of 1st Test Trial		F Ratio	Level of Significance
	Mean	DF	Mean	DF		
M6-6	25.64	22	82	103.18	2.20	5%
M6-21	41.18	22	86.18	96.06	2.09	5%
M21-6	12.58	24	57.08	34.01	4.54	1%
M21-21	7.92	24	51.33	42.86	6.48	1%

TABLE 20

Comparing the time in seconds of the first test trial of the four experimental groups. (Experiment 1.)

Group	Mean	F Ratio	DF	Level of Significance
M 6- 6	82	1.051	103.1844	
M 6-21	86.182		96.0586	
M 6- 6	82	1.4365	103.1844	20% level
M 21- 6	57.083		34.0056	
M 6-21	86.182	1.6789	96.0586	5% level
M 21-21	51.333		42.8640	
M 21- 6	57.083	1.1120	34.0056	
M 21-21	51.333		42.8640	

TABLE 21

Indicating critical ratios of the differences between entries of the two experimental groups into the six alleys on the first test trial. (Experiment 2.)

Groups	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
Lm6-21 LM21-21	.30	1.06	1.06	1.06	1.01	.21

TABLE 22

Indicating the critical ratios obtained by comparing the percent of entries into each alley with the percent of entries into each of the other alleys for the animals in Group LM6-21 on the first test trial. (Experiment 2.)

	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
Alley 1	--	1.10	1.10	1.10	.68	1.31
Alley 2	--	--	0	0	1.73	2.45
Alley 3	--	--	--	0	1.73	2.45
Alley 4	--	--	--	--	1.73	2.45
Alley 5	--	--	--	--	--	.59

TABLE 23

Indicating the critical ratios obtained by comparing the percent of entries into each alley with the percent of entries into each of the other alleys for the animals in Group LM21-21 on the first test trial. (Experiment 2.)

	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
Alley 1	--	0	0	0	0	1.70
Alley 2	--	--	0	0	0	1.70
Alley 3	--	--	--	0	0	1.70
Alley 4	--	--	--	--	0	1.70
Alley 5	--	--	--	--	--	1.70

TABLE 24

Indicating the critical ratios obtained by comparing the percent of entries into each alley with the percent of entries into each of the other alleys for the total number of animals in Experiment 2 on the first test trial.

	Alley 1	Alley 2	Alley 3	Alley 4	Alley 5	Alley 6
Alley 1	--	.61	.61	.61	.49	2.14
Alley 2	--	--	0	0	1.10	2.78
Alley 3	--	--	--	0	1.10	2.78
Alley 4	--	--	--	--	1.10	2.78
Alley 5	--	--	--	--	--	1.62

TABLE 25

Showing the significance between the number of responses to the correct side (responses to alleys 1, 2, and 3) made by the two experimental groups in 20 trials. (Experiment 2)

Group	Mean	F Ratio	DF	Level of Significance
LM 6-21	14.333	1.0157	472.068	
LM 21-21	14.111		2502.378	

TABLE 26

Indicating the significance between comparable groups in Experiments I and II in the number of times the correct side (alleys 1, 2, and 3) was chosen in 20 trials.

Group		Mean	F Ratio	DF	Level of Significance
M	6-21	11.091	1.2923	91.6212	
LM	6-21	14.333		472.068	
M	21-21	13.333	1.0584	380.2608	
LM	21-21	14.111		2502.378	

TABLE 27

Indicating the significance obtained by comparing the number of errors made by the two experimental groups to reach 20 reinforcements in the training trials. (Experiment 2)

Groups	Mean	F Ratio	DF	Level of Significance
LM 6-21	19.5	1.9942	28.5936	5% level
LM 21-21	9.778		53.8632	

TABLE 28

Showing the significance between the comparable groups in Experiments 1 and 2 in the number of errors made before reaching 20 reinforcements in the training trials.

Group		Mean	F Ratio	DF	Level of Significance
LM	6-21	19.5	2.2579	28.5936	1%
M	6-21	8.636		70.9060	
LM	21-21	9.778	1.0115	53.8632	
M	21-21	9.667		30.6312	

TABLE 29

Comparing the time in seconds taken by the two experimental groups to make the first correct response for the last day of the training trials.
(Experiment 2.)

Group	Mean	F Ratio	DF	Level of Significance
LM 6-21	17.667	1.8275	12.8028	20%
LM 21-21	9.667		41.8194	

TABLE 30

Comparing the mean time in seconds taken on the first correct choice on the last day of the training trials with the mean time taken on the first test trial for the two experimental groups. (Experiment 2)

Group	Time of 1st Correct Trial on last day of Training Trials		Time of 1st Test Trial		F Ratio	Level of Significance
	Mean	DF	Mean	DF		
LM6-21	17.67	12.80	56.83	17.68	3.22	5%
LM21-21	9.67	41.82	30	56.47	3.10	1%

TABLE 31

Comparing the time in seconds of the first test trial of the two experimental groups.
(Experiment 2)

Group	Mean	F Ratio	DF	Level of Significance
LM 6-21	56.833	1.8944	17.6856	5%
LM 21-21	30		56.4678	

TABLE 32

Comparing the number of correct responses in 20 test trials made by the animals in Group LM6-21 having the regular amount of training and those having extra runs in the lighted alley.

Group	Regular Training		Extra Training		F Ratio	Level of Significance
	Mean	DF	Mean	DF		
LM6-21	6.83	34.72	12.5	45.87	1.83	5%

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